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REPORT

Hospital queue management dashboard

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Introduction

A Hospital Queue Management System is a digital solution designed to replace the traditional, chaotic queuing process in hospital OPDs. It automates the patient flow using a token-based system, significantly reducing wait times, managing crowds, and enhancing the experience for both patients and hospital staff.

Objectives

- 1.To digitize and automate the patient queuing process.
- 2.To minimize patient wait times and reduce physical crowding.
- 3.To provide real-time queue status to patients via display screens.
- 4.To increase administrative efficiency and staff productivity.

Problem Statement

1. Manual patient queues in hospitals are inefficient, leading to long waits, confusion, and patient dissatisfaction.

2. There is a pressing need for an organized, transparent, and fair system to manage patient flow effectively.

3. We propose a software-based queue management system that issues digital tokens and provides live updates, ensuring an orderly and efficient process.

Methodology:-

1.Requirement Analysis: Understanding the problems with the current manual system.

2.System Design: Creating the workflow and algorithm.

3.Implementation: Coding the core logic using Python.

4.Testing: Verifying the system with sample data.

5.Deployment Plan: Outlining how the system can be deployed in a real-world setting.

code:-

```
class HospitalQueue: def __init__(self,
                                department):

    self.queue = [] self.department =
    department self.current_token = 0

def generate_token(self, patient_name):

    self.current_token += 1
    self.queue.append((self.current_token,
                        patient_name))

    print(f"Token #{self.current_token}
    generated for {patient_name}.")

    def call_next(self):

        if self.queue:

            token, patient = self.queue.pop(0) print(f"-->
            Token #{token}, {patient},
            please proceed to {self.department}.")

        else:

            print("No patients in the queue.")

# Demo
cardio_queue = HospitalQueue("Cardiology")
cardio_queue.generate_token("Aditya Ghanekar")
cardio_queue.generatetoken("Omkar Hambire")
```

Application:-

- Hospital OPDs (Cardiology, Neurology, etc.)
- Diagnostic Centers and Blood Test Labs. Pharmacy
- Counters for medicine collection.
- Hospital Billing and Admission Desks.

Result o/p:-

The implementation of the Hospital Queue Management System was successful. The Python code effectively simulated a real-world queuing process for a hospital department. The system correctly generated unique, sequential tokens for patients as they checked in, placing them into a digital queue. It then processed the queue in a strict First-In-First-Out (FIFO) order, calling each patient by their token number and name.

HOSPITAL QUEUE MANAGEMENT SYSTEM

Queue No	Patient Name	Condition	Emergency	Status	ETA	Seriousness	Symptoms
2	MEHUL	Internal Damage - Emergency	Yes	Waiting	5 min	● High	HEART BREAK
1	OMKAR	Physical Damage - Emergency	No	Waiting	15 min	● Low	HAND SLIP

+ Add New Patient - Describe Symptoms and Set Seriousness

Patient Name:

☒ Internal Normal ☐ Physical Emergency ☒ Physical Normal

☒ Critical Emergency ☐ Internal Emergency

Describe symptoms for Physical Damage - Normal:

Seriousness Level (1=Critical, 5=Minimal)

☐ 1 - Critical ☒ 2 - High ☐ 3 - Medium ☐ 4 - Low ☒ 5 - Minimal

OK Cancel

CURRENT PATIENT DETAILS

No patient currently in consultation.

Please call next patient to start consultation.

+ Add Patient ← Call Next ✓ Complete Current ✕ Remove ↺ Refresh Priority Sort View Full Symptoms

📍 Total: 2 | ⌚ Waiting: 2 | 🚑 In Consultation: 0 | ✓ Completed: 0 | 🚑 Emergency: 1 | ● Critical: 0 | 🚑 Current: None

Conclusion:-

- The Hospital Queue Management System successfully addresses the inefficiencies of manual patient queuing. By implementing a digital, token-based FIFO queue, the project demonstrates a clear path towards reducing wait times, eliminating confusion, and creating a more streamlined and stress-free environment for patients and healthcare providers alike. The system is scalable and can be further enhanced with features like online appointments and SMS notifications.